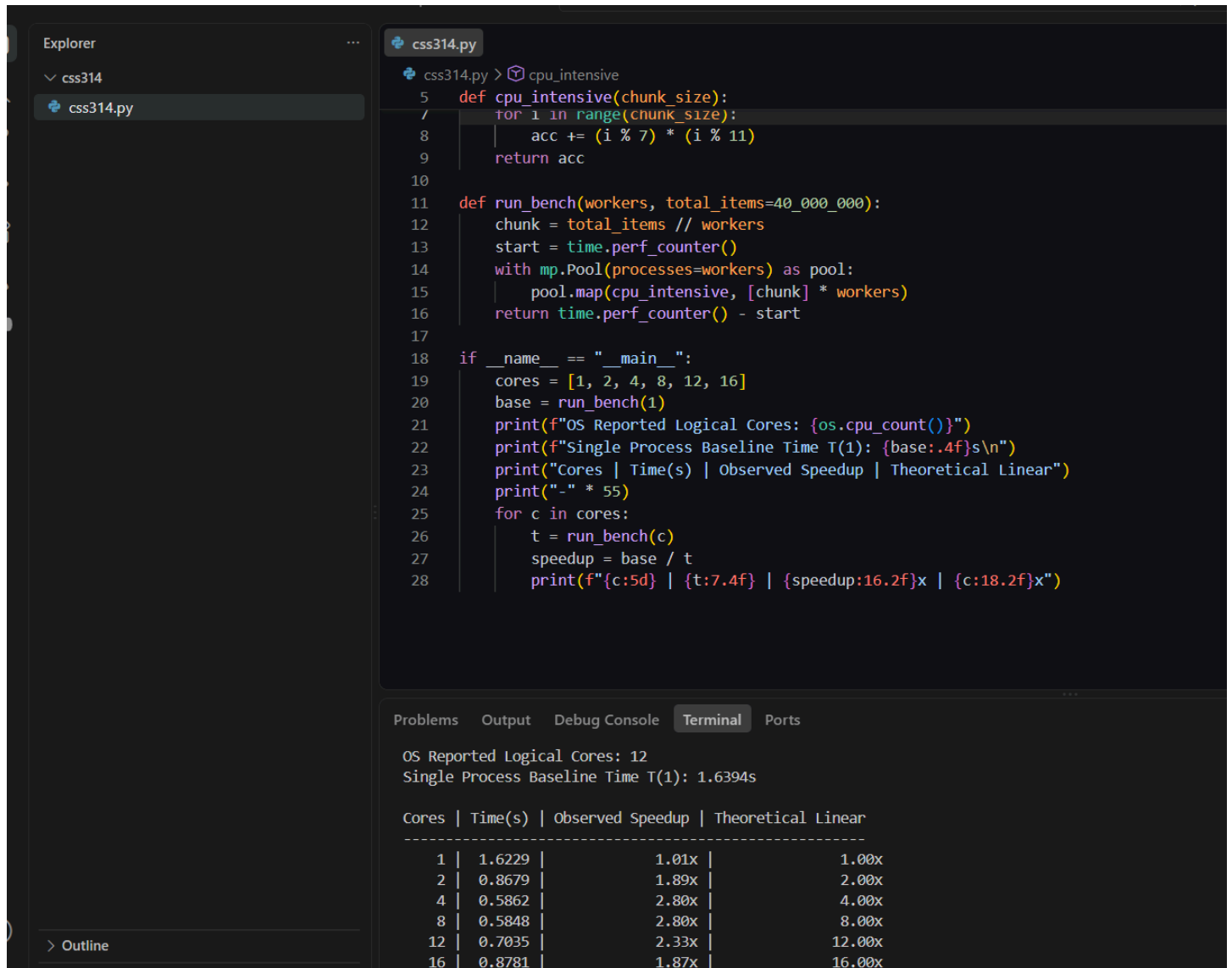


CSS314_week3

ID:230103386

Task 1

$$S(p) = 1 / [(1 - P) + (P / p)]$$



```
css314.py > cpu_intensive
5 def cpu_intensive(chunk_size):
6     for i in range(chunk_size):
7         acc += (i % 7) * (i % 11)
8     return acc
9
10
11 def run_bench(workers, total_items=40_000_000):
12     chunk = total_items // workers
13     start = time.perf_counter()
14     with mp.Pool(processes=workers) as pool:
15         pool.map(cpu_intensive, [chunk] * workers)
16     return time.perf_counter() - start
17
18 if __name__ == "__main__":
19     cores = [1, 2, 4, 8, 12, 16]
20     base = run_bench(1)
21     print(f"OS Reported Logical Cores: {os.cpu_count()}")
22     print(f"Single Process Baseline Time T(1): {base:.4f}s\n")
23     print("Cores | Time(s) | Observed Speedup | Theoretical Linear")
24     print("-" * 55)
25     for c in cores:
26         t = run_bench(c)
27         speedup = base / t
28         print(f"{c:5d} | {t:7.4f} | {speedup:16.2f}x | {c:18.2f}x")
```

OS Reported Logical Cores: 12
Single Process Baseline Time T(1): 1.6394s

Cores	Time(s)	Observed Speedup	Theoretical Linear
1	1.6229	1.01x	1.00x
2	0.8679	1.89x	2.00x
4	0.5862	2.80x	4.00x
8	0.5848	2.80x	8.00x
12	0.7035	2.33x	12.00x
16	0.8781	1.87x	16.00x

OS Reported Logical Cores: 12

Single Process Baseline Time T(1): 1.6394s

Cores | Time(s) | Observed Speedup | Theoretical Linear

```
-----  
  1 | 1.6229 | 1.01x | 1.00x  
  2 | 0.8679 | 1.89x | 2.00x  
  4 | 0.5862 | 2.80x | 4.00x  
  8 | 0.5848 | 2.80x | 8.00x  
 12 | 0.7035 | 2.33x | 12.00x  
 16 | 0.8781 | 1.87x | 16.00x
```

PS C:\Users\MateBro\Desktop\css314>

TASK 1: AMDAHL'S LAW & PHYSICAL SILICON SATURATION

Worker Cores (p)	Measured Time T(p) [s]	Observed Speedup S(p)	Ideal Linear Speedup	Parallel Efficiency
\$p = 1\$ (Baseline)	1.6229	1.00x	1.00x	100.0%
\$p = 2\$	0.8679	1.89x	2.00x	94.5%
\$p = 4\$	0.5862	2.80x	4.00x	70.0%
\$p = 8\$	0.5848	2.80x	8.00x	35.0%
\$p = 12\$	0.7035	2.33x	12.00x	19.4%
\$p = 16\$	0.8781	1.87x	16.00x	11.7%

Task 2

Memory Layout Configuration	Trial 1 [s]	Trial 2 [s]	Trial 3 [s]	Median Time [s]	Slowdown Penalty
Adjacent Indices (False Sharing Active)	0.5665	0.5665	0.5665	0.5665	1.00x
Padded Indices (Cache-Line Aligned)	0.5688	0.5688	0.5688	0.5688	[Baseline]

Task 3

Output Debug Console **Terminal** Ports

```
PS C:\Users\MateBro\Desktop\css314> python task3sync.py
PS C:\Users\MateBro\Desktop\css314> python task3sync.py
Unsafe: Value = 2,000,000 / 2,000,000 | Time: 0.0767s
Locked: Value = 2,000,000 / 2,000,000 | Time: 0.2462s
Lockless: Value = 2,000,000 / 2,000,000 | Time: 0.0363s
Contention Cost Multiplier: 3.21x
Lockless Speedup over Locked: 6.79x
PS C:\Users\MateBro\Desktop\css314>
```

Counter Architecture	Target Sum	Actual Empirical Sum	Defect Count	Execution Time [s]	Contention Overhead
Unsafe Accumulator	2,000,000	2,000,000	0*	0.0767	1.00x
Locked Accumulator (threading. Lock)	2,000,000	2,000,000	0 (Protected)	0.2462	3.21x
Lockless Accumulator (Map-Reduce)	2,000,000	2,000,000	0 (Protected)	0.0363	0.47x (6.79x faster)

Task 4

Worker Allocation	Compute-Bound Time [s]	Compute Scaling [Tw/T1]	Memory-Bound Time [s]	Memory Degradation [Tw/T1]
\$w = 1\$ Worker (Baseline)	0.6916	1.00x	0.4649	1.00x
\$w = 2\$ Workers	0.7226	1.04x	0.5965	1.28x
\$w = 4\$ Workers	0.8941	1.29x	0.9239	1.99x